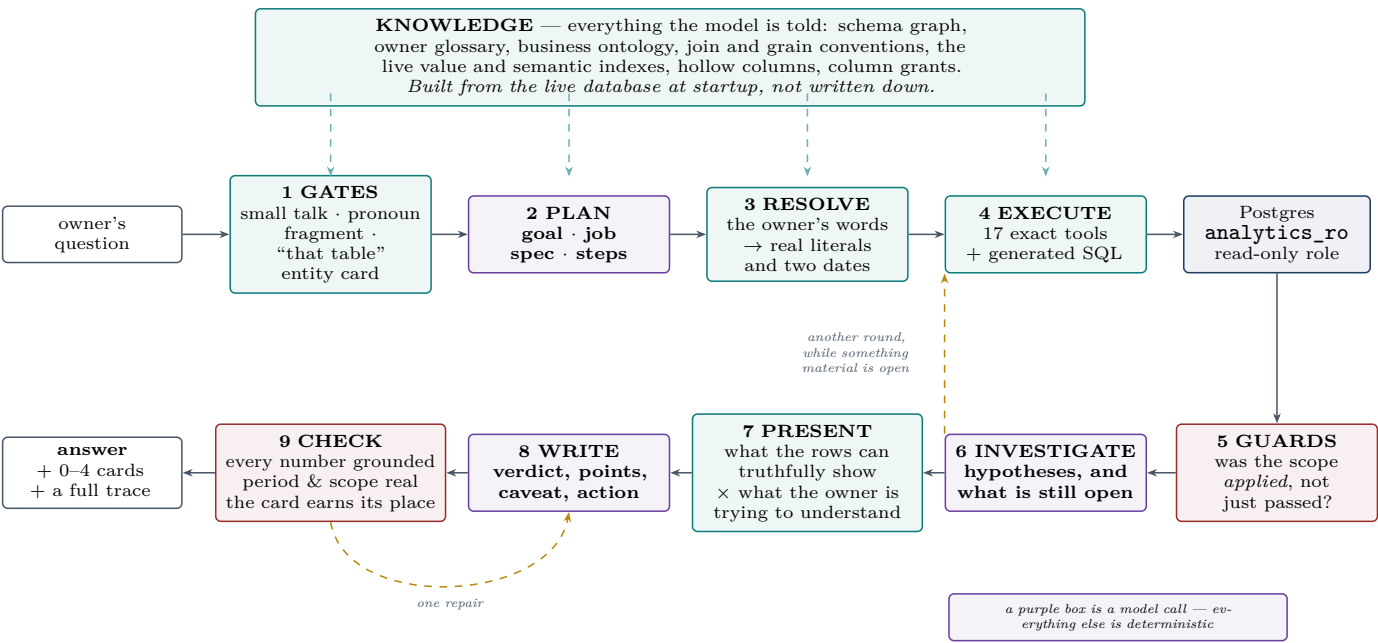


An AI analytics layer over the live clinical Postgres. The owner types a question in their own words — English, Hinglish, or a four-word fragment — and Pulse plans an analysis, runs it, decides for itself whether it has learned enough, and writes an answer in which every figure traces back to a step that produced it.

How one question flows



Cost of that: 3 model calls and 4–6 seconds for a figure. A real investigation is a median of 7 calls and 22–32 seconds, with a hard ceiling of 180s that the answer must *disclose* if it hits.

What each layer is responsible for

Layer	Decides	The rule it exists to enforce
Gates	is this even an analysis question	“In that table” is a <i>conversation</i> problem, not an entity lookup. A pronoun with nothing to point at is a question, not a search.
Knowledge	everything the model is told	Built from the live database at startup, not written down. The <i>only</i> lever measured to move accuracy on this workload.
Plan	goal, analytical job, scope, period, steps	The analyst writes a <i>contract</i> before any SQL exists — the owner’s word next to the resolved literal.
Resolve	what the owner’s words mean here	Resolved against the owner’s own sentence, never the planner’s paraphrase. “CT” is a modality here, not the lab code for Clotting Time.
Execute	17 exact tools, or generated SQL	A registry tool is free and repeatable; query is general. A close number is a wrong number, so a qualifier the tools cannot express goes to query .
Guards	may this figure be used at all	A tool that applied a scope <i>says so</i> . Silence is a refusal — a wrong number with a confident label is the worst thing this system can do.
Investigate	is anything material still open	Continuing is a consequence of an unresolved claim, <i>not</i> of the model’s appetite. Stopping always has a stated reason.
Present	what may be drawn, and how much goes in prose	The <i>rows</i> decide what a card can truthfully show; the <i>job</i> decides which of those is useful. No words are matched against the question.
Check	may this answer ship	Every figure must come from a step or follow from two that do. An unclosed investigation must say so in the verdict, not a trailing caveat.

THE SECURITY BOUNDARY

Pulse reads through **analytics_ro**: a Postgres role with **SELECT** on a whitelist of tables, *column-level* grants on **Patient** and **User**, read-only transactions and an eight-second statement timeout. **The role is the real boundary**. A deterministic validator sits on top of it and blocks writes, secrets, implicit cross products, fan-out double counting and unbounded **SELECT ***.

Whether a result may *identify a person* is a separate question, decided at the route from the asker’s role and passed in as policy — never inferred from the shape of a query. The surface is owner-only, and every question is audited.

Sixteen days, eleven architectures, roughly 9,000 measured model calls. Two results did all the shaping.

- 1. Ground, never verify.**

Three post-hoc verification mechanisms were built and measured: 3-sample execution voting (**22.0 vs 22.0** at 3× the tokens), LLM inspection of finished answers (**no** score change), and an automatic repair loop (**0** genuine errors fixed over three runs).

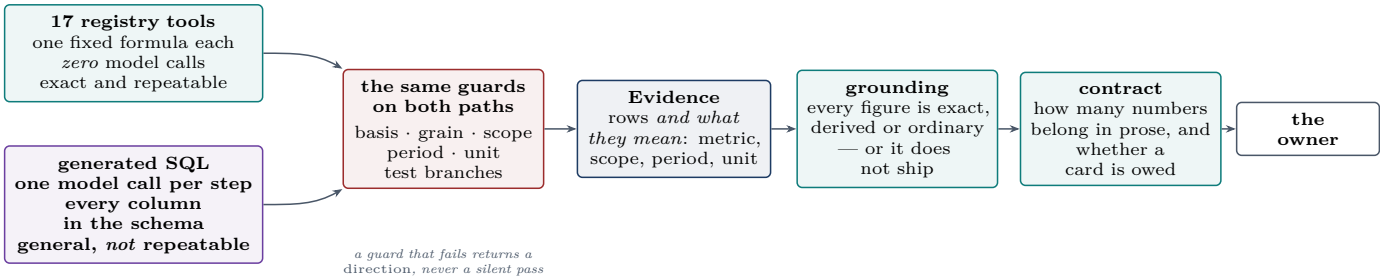
Three mechanisms measuring zero is one finding, not three disappointments: *the errors are systematic*. Sampling wrong reasoning three times returns the same wrong answer three times. So every token is spent on what the model is **told**, never on checking what it **said**.
- 2. An invariant described in a prompt is not an invariant.**

Every rule this system enforces was first written as an instruction to the model. Every one was then violated, in production, on a question that mattered.

“Never attach a card that only repeats a single figure” — attached one on 3 of 6. *“A due is the computed balance, never the status flag”* — written into the conventions three ways, ignored each time.

The rules that survive are the ones that became **code**.

The two execution paths, and the wall between evidence and the owner



Seventeen tools are deterministic and free, so a plan can look at six things before deciding what matters. Only **query** buys latency — and only **query** can give two different answers to the same question, which is why the guards sit on both paths and not just on it.

WHAT IT COSTS

a figure	3 calls, 4–6s
an investigation	median 7 calls, 22–32s
worst observed	~114s
hard ceiling	180s, and it must say so
per question	~8k tokens in, ~1k out
full benchmark	under \$2

WHAT IS TESTED

pure layers (free)	115 / 115
deterministic core (free)	65 / 65
the judge itself (free)	12 / 12
frontend resolution (free)	7 / 7
end-to-end chain (paid)	12 / 12
adversarial (paid)	28 / 29

The paid ones matter most: they test that the gears stay connected, not that each one turns.

THE THREE THINGS STILL WRONG WITH IT

1. Grain is enforced by name, not carried as data. Four money failures are each held back by a *specifically named* repair. A fifth mismatch would not be caught.

2. Scope is still verified backwards. Time is bound forward into the query; scope is re-derived from English and then recognised by a regex table — the one structure this design exists to delete.

3. No degraded mode. When the model is unreachable there is no answer at all, even for questions that need no model.

- THE BAR, IN ORDER OF WHAT IT COSTS THE OWNER**
1. Never present a figure for a wider question than was asked — a centre-wide total labelled “Chintal” is the worst output this system can produce. 2. Never state a number it cannot trace to a step. 3. Never report a limitation of Pulse as a fact about the business. 4. Never present an incomplete investigation as settled. 5. Never attach a card that adds nothing — noise everywhere is the same as nothing anywhere. 6. Prefer silence to a confident wrong number; silence is recoverable.